

SOEMS Test Portal — Sample Questions

General Aptitude & CS Basics — for portal testing purposes

Section A: Standard Questions (MCQ)

1. What is the time complexity of binary search on a sorted array?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(n^2)$
- D) $O(1)$

Correct answer: B

2. Which HTTP status code indicates a successful POST request that created a resource?

- A) 200
- B) 201
- C) 301
- D) 404

Correct answer: B

3. In React, which hook is used to manage local component state?

- A) `useEffect`
- B) `useContext`
- C) `useState`
- D) `useMemo`

Correct answer: C

4. What does SQL stand for?

- A) Structured Query Language
- B) Simple Query Logic
- C) Sequential Query Language
- D) Server Query Language

Correct answer: A

5. Which of these is NOT a valid HTTP method?

- A) GET
- B) PUSH
- C) DELETE
- D) PATCH

Correct answer: B (invalid — the real method is PUT)

6. What is the capital of Australia?

- A) Sydney
- B) Melbourne

C) Canberra

D) Perth

Correct answer: C

7. Which data structure uses FIFO (First In, First Out) ordering?

A) Stack

B) Queue

C) Tree

D) Graph

Correct answer: B

8. In a relational database, what does a "foreign key" do?

A) Encrypts data

B) Links a row to a primary key in another table

C) Deletes duplicate rows

D) Indexes text search

Correct answer: B

9. What is the output of $2 ** 3$ in Python?

A) 6

B) 8

C) 9

D) Error

Correct answer: B

10. Which of these is a NoSQL database?

A) PostgreSQL

B) MySQL

C) MongoDB

D) SQLite

Correct answer: C

Section B: Edge-Case Questions (UI & Proctoring Stress Tests)

11. Long-text stress test

Use a question with a 300+ word passage before the actual question text, to verify ExamPage.jsx scrolls and renders correctly without breaking the layout or cutting off content on smaller screens.

12. Special-character test

What does the operator `&&` mean in JavaScript, and how does it differ from `||`, `!`, and `===`?

Tests that quotes, ampersands, and symbols render and save correctly (no HTML-escaping bugs).

13. Minimal-content test

2 + 2 = ? (Options: 3 / 4 / 5 / 22)

Tests minimal-content rendering.

14. Negative-marking / grading-bias test

Design a question where the correct answer is intentionally the last option (D), to verify the grading logic isn't silently defaulting to option A.

15. Timer-boundary test

Place this as the last question in a 2-minute mock exam to verify the timer auto-submits correctly when the clock hits zero mid-question.

Tip: Import Section A as a standard exam to test scoring and the violation/kick pipeline. Use Section B individually to stress-test specific parts of the UI and proctoring logic.